

Lab sheet-02

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// TASK 1: Create product array
//=====
const products = [
  {
    id: 1,
    name: "Headphones",
    price: 2499,
    image: "images/headphones.jpg"
  },
  {
    id: 2,
    name: "Smart Watch",
    price: 3999,
    image: "images/watch.jpg"
  },
  {
    id: 3,
    name: "Bluetooth Speaker",
    price: 1999,
    image: "images/speaker.jpg"
  },
  {
    id: 4,
    name: "Keyboard",
    price: 3499,
    image: "images/keyboard.jpg"
  },
  {
    id: 5,
    name: "Gaming Mouse",
    price: 1499,
    image: "images/mouse.jpg"
  },
  {
    id: 6,
    name: "USB Charger",
    price: 999,
    image: "images/charger.jpg"
  },
  {
    id: 7,
    name: "Laptop Stand",
    price: 1299,
    image: "images/stand.jpg"
  },
  {
    id: 8,
    name: "Webcam",
    price: 2299,
    image: "images/webcam.jpg"
  }
];
// TASK 3: Create cart and use localStorage
//=====
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let cart = JSON.parse(localStorage.getItem("cart")) || []; function
saveCart() {
    localStorage.setItem("cart", JSON.stringify(cart));
}

// TASK 8: Update cart item count
// =====
function updateCartCount() {

    const cartCount = document.getElementById("cart-count");

    if (cartCount) {

        const count = cart.reduce(function(total, item) {
            return total + item.quantity;
        }, 0);

        cartCount.textContent = count;
    }
}

// TASK 2: Dynamically display products
// =====
function displayProducts() {
    const productList = document.getElementById("product-list");
    if (!productList) return;
    productList.innerHTML = "";
    products.forEach(function(product) {
        const card = document.createElement("div");
        card.innerHTML = `
            
            <h3>${product.name}</h3>
            <p>₹${product.price}</p>
            <button onclick="addToCart('${product.id}')">
                Add to Cart
            </button>`;
        productList.appendChild(card);
    });
}

// TASK 3: Add product to cart
// =====
function addToCart(id) {
    const product = products.find(function(product) {
        return product.id === id;
    });
    const existingItem = cart.find(function(item) {
        return item.id === id;
    });
    if (existingItem) {
        existingItem.quantity++;
    } else {
        cart.push({
            id: product.id,
            name:
            product.name,
            price:
            product.price,
            image:
            product.image,
            quantity: 1
        });
    }
    saveCart();
    updateCartCount();
    alert("Product added to cart!");
}

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}

// TASK 4: Display cart items
// =====
function displayCart() {
    const cartItems = document.getElementById("cart-items");
    if (!cartItems) return; cartItems.innerHTML = ""; if
    (cart.length === 0) {
        cartItems.innerHTML = "<p>Your cart is empty.</p>";
        calculateTotal();
    }
    return;
    cart.forEach(function(item) { const div =
    document.createElement("div");
    div.innerHTML =
    <h3>${item.name}</h3> <p>Price:
    ₹${item.price}</p>
    <label> Quantity:
    <input type="number"
    min="1"
    value="${item.quantity}"
    onchange="changeQuantity(${item.id}, this.value)"
    >
    </label>

    <button onclick="removeFromCart(${item.id})">
    Remove
    </button>
    <hr>
    cartItems.appendChild(div);
    });
    calculateTotal();
}

// TASK 5: Change quantity
// =====
function changeQuantity(id, quantity) {
    const item = cart.find(function(item) {
        return item.id === id;
    }); if
    (item) {

        item.quantity = Number(quantity);

        if (item.quantity < 1) {
            item.quantity = 1;
        }
        saveCart();
    }
    displayCart();
    updateCartCount();
}

// TASK 6: Remove item from cart
// =====
function removeFromCart(id) { cart = cart.filter(function(item) {
    return item.id !== id;
});
    saveCart();
    displayCart();
    updateCartCount();
}

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}
// TASK 5: Calculate grand total using reduce()
// =====
function calculateTotal() {
  const totalElement = document.getElementById("cart-total");

  if (!totalElement) return;

  const total = cart.reduce(function(sum, item) {

    return sum + item.price * item.quantity;

  }, 0);

  totalElement.textContent = "₹" + total; }

// TASK 7: Checkout form validation
// =====
function setupCheckout() {
  const form = document.getElementById("checkout-form");
  if (!form) return;
  form.addEventListener("submit", function(event) {
    event.preventDefault();
    const name = document.getElementById("name").value.trim();
    const address =
      document.getElementById("address").value.trim();
    const pincode =
      document.getElementById("pincode").value.trim();
    const phone =
      document.getElementById("phone").value.trim();
    // Clear old errors
    document.getElementById("name-error").textContent = "";
    document.getElementById("address-error").textContent = "";
    document.getElementById("pincode-error").textContent = "";
    document.getElementById("phone-error").textContent = "";
    let valid = true; // Name validation if (name === "") {
      document.getElementById("name-error").textContent =
        "Name is required";
    valid = false;
  }
  // Address validation
  if (address === "") {
    document.getElementById("address-error").textContent =
      "Address is required";
    valid = false;
  }
  // Pincode validation
  if (!/^d{6}$/.test(pincode)) {

    document.getElementById("pincode-error").textContent =
      "Pincode must be exactly 6 digits";
    valid = false;
  }
  // Phone validation if
  (!/^d{10}$/.test(phone)) {
    document.getElementById("phone-error").textContent =
      "Phone must be exactly 10 digits";
    valid = false;
  }
}

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    }
    // TASK 8: Successful order
    // =====
    if (valid) {
        document.getElementById("confirmation").textContent =
            "Order placed successfully!";
        cart = [];
        localStorage.removeItem("cart");
        updateCartCount();      form.reset();
    }
    });
}
// RUN ALL FUNCTIONS
// =====
document.addEventListener("DOMContentLoaded", function() {

    displayProducts();

    displayCart();

    updateCartCount();

    setupCheckout();

});

```

